When Does Detect-and-Reweight Work? Epoch Confounds and Adaptive Reweighting in LM Pretraining

Anonymous ACL submission

Abstract

Detect-and-reweight methods suppress likely-synthetic samples during pretraining via $w(x) = (1 - q(x))^b$, where $q(x)$ is a detector's synthetic probability. Two questions are open: does hard filtering (Detect and Remove, D&R) outperform soft reweighting, and how should b be set without per-model grid search?

We address both through controlled experiments in a self-contamination setting: Qwen3-0.6B continued pretraining on 132M FineWeb tokens with 40% synthetic contamination (3 seeds). First, D&R's reported advantage is entirely a multi-epoch artifact: at equal content epochs, reweighting and D&R differ by only 0.006 perplexity (PPL; N=3). Second, the in-distribution PPL improvement from adaptive reweighting does not transfer to out-of-distribution corpora (OpenWebText, Wikipedia), consistent with distribution matching rather than general quality gain. Third, a dual-constrained adaptive b^* improves over uniform weighting on in-distribution evaluation with directional consistency across all seeds (95% confidence interval (CI) of the adaptive-to-grid gap includes possible degradation up to 0.10 PPL); our results suggest detector signal type may matter more than discrimination accuracy (3 detectors, single seed).

The method fails at low contamination (ρ=0.2), which establishes a clear applicability boundary. N=3 precludes formal significance testing; we report effect sizes and directional consistency.

1 Introduction

The detect-and-reweight paradigm (Drayson et al., 2025) has emerged as a response to synthetic contamination in web corpora, now estimated at 20–60% of crawled data (Cheng et al., 2025). Training on such data risks model collapse (Shumailov et al., 2024; Dohmatob et al., 2025), motivating methods that downweight likely-synthetic samples via $w(x) = (1 - q(x))^b$, where $q(x)$ is a detector's synthetic probability. An alternative, Detect and Remove (D&R), hard-filters samples above a threshold (Feng et al., 2025).

Despite growing adoption, fundamental questions remain. Does D&R genuinely outperform soft reweighting, or do published comparisons confound filtering quality with multi-epoch exposure? How should b be chosen, given that Drayson et al. (2025) find optimal values vary from b=10 (GPT-2 124M) to b=1 (SmolLM 1.7B), which requires costly grid search? What detector properties are necessary for effective reweighting?

We address these through controlled experiments on Qwen3-0.6B continued pretraining (132M FineWeb tokens, ρ=0.4, 3 seeds), with extensions to $\rho \in \{0.2, 0.6\}$ and 1.7B parameters. This scope, continued pretraining on moderate-scale corpora with known contamination (Gururangan et al., 2020), does not address from-scratch pretraining at web scale. We formulate adaptive exponent selection as a dual-constrained problem: find the minimum b such that the weighted effective real proportion $\alpha_{\text{eff}}(b) \geq \tau$ (quality floor, motivated by convergence theory (Wang et al., 2026)), subject to Kish's effective sample size $\text{ESS}(b) \geq n_{\min}$ (stability ceiling) (Kish, 1965; Vehtari et al., 2024).

Our contributions identify two evaluation confounds and characterize adaptive reweighting:

- **Epoch confound.** D&R's PPL advantage is entirely attributable to multi-epoch exposure; epoch-matched controls (N=3) reduce the gap to Δ=0.006 (Table 1).
- **Distribution matching confound.** The in-distribution improvement (Δ=−0.043 on FineWeb) does not transfer out-of-distribution (OpenWebText +0.06, Wikipedia +0.14; §5.2).
- **Mechanism analysis.** The α_{eff} quality constraint binds in 38.2% of batches at 4.9× lower exponents than ESS-only (§5.4); two generative detectors reverse the advantage; signal type may

matter more than AUC (3 detectors, single seed; §5.2). The method reduces b grid search to a single default τ ($\Delta{=}-0.043$, directionally consistent across all seeds; gap to grid-best 0.031, 95% CI $[-0.039, 0.101]$), but fails at $\rho{=}0.2$ and shows no effect at 1.7B.

2 Related Work

Synthetic Contamination and Model Collapse. Training on recursively generated data leads to irreversible quality degradation (*model collapse*) (Shumailov et al., 2024). Alemohammad et al. (2024) formalize this as Model Autophagy Disorder and show that quality or diversity inevitably degrades without fresh real data, even when original training data is retained. More critically, Dohmatob et al. (2025) prove under scaling-law regimes that even a vanishingly small synthetic fraction (e.g., $1/1000$) suffices to trigger collapse, ruling out data scaling as a remedy; Dohmatob et al. (2024) characterize this as a change of scaling laws. Gerstgrasser et al. (2024) show that accumulating real data alongside synthetic data can break the recursion, though at substantial collection cost. Kang et al. (2025) systematically study scaling laws for synthetic data in pretraining, finding that rephrased synthetic data mixed with natural text improves data efficiency but textbook-style data degrades loss on many domains. Cheng et al. (2025) survey the broader contamination landscape and confirm that synthetic content now constitutes an estimated 20–60% of web corpora. As language models are deployed at scale for content generation (Radford et al., 2019; Brown et al., 2020; OpenAI, 2023), these results motivate *contamination-aware* training strategies that explicitly account for synthetic provenance rather than treating all tokens equally.

Detection and Filtering of Machine-Generated Text. Given the severity of contamination, a natural response is to detect and remove synthetic content. DetectGPT (Mitchell et al., 2023) performs zero-shot detection via probability curvature; Fast-DetectGPT (Bao et al., 2024) accelerates this with conditional probability curvature; Binoculars (Hans et al., 2024) uses cross-perplexity ratios between two models; Min-k% Prob (Shi et al., 2024) uses the average log-probability of the lowest-k% tokens. All produce continuous confidence scores suitable as input to downstream weighting, including ours. Kirchenbauer et al. (2023) propose watermarking as a complementary provenance mechanism. On the mitigation side, Feng et al. (2025) prove that even an imperfect verifier prevents collapse, but apply hard binary filtering that discards borderline samples; Yi et al. (2025) extend this with convergence guarantees for verified synthetic data. Zibakhsh Shabgahi et al. (2025) propose truncated cross-entropy loss to implicitly discount high-confidence (likely synthetic) tokens during training. Hard filtering requires a precision–recall threshold; soft reweighting avoids this trade-off by assigning continuous weights.

Most directly related to our work, Drayson et al. (2025) propose $w_j = (1{-}q_j)^b$ for collapse prevention via detect-and-reweight, where q_j is the detector's synthetic probability. We adopt the same weighting formula but differ in several fundamental ways. **(i)** Drayson et al. select b via validation-set grid search and fix it throughout training; different model scales require different values ($b{=}10$ for $\leq$360M, $b{=}1$ for 1.7B), so the choice is *per-model* and expensive. Our b^* is derived per-batch from dual constraints ($\alpha_{\text{eff}} \geq \tau$, $\text{ESS} \geq n_{\min}$) at zero validation cost. **(ii)** Their SIR resampling discretizes the weight distribution; we apply direct loss reweighting with continuous gradients. **(iii)** They employ a ModernBERT-150M detector (AUC = 0.986) on WikiText-2 ($\sim$2M tokens); we show that a lightweight TF-IDF+LR detector (AUC = 0.939) suffices on FineWeb-scale data (132M+ tokens), and score calibration may matter more than raw detection accuracy.

Domain-Level Data Mixing. A parallel line of work optimizes data composition without targeting synthetic contamination. DoReMi (Xie et al., 2023a) learns mixture weights via Group DRO (Sagawa et al., 2020) with a proxy model, DSIR (Xie et al., 2023b) performs importance resampling with n-gram features, and Ye et al. (2025) discover predictable functional forms relating mixing ratios to performance. AIOLI (Chen et al., 2025) unifies these into an online adaptive framework, while Yang et al. (2026) train an RL agent for domain-weight selection. DataComp-LM (Gadre et al., 2024) and OLMo (Groeneveld et al., 2024) provide standardized benchmarks and open infrastructure for data curation at scale. Continued pretraining (Gururangan et al., 2020) adapts models to specific domains, a setting where contamination-aware methods are directly applicable. These domain-level approaches do not address instance-level contamination within a domain.

Instance-Level Data Selection and Reweighting. Deduplication at the corpus level (Lee et al., 2022) is a prerequisite; data-constrained regimes (Muennighoff et al., 2024) and data quality filtering (Longpre et al., 2024) further shape training distributions. SemDeDup (Abbas et al., 2023) deduplicates via embedding similarity, SoftDedup (He et al., 2024) downweights high-frequency samples via n-gram commonness, and QuRating (Wettig et al., 2024) selects high-quality instances via learned quality scores. Curriculum learning (Bengio et al., 2009) and focal loss (Lin et al., 2017) upweight examples by difficulty, Sow et al. (2025) derive sample weights from per-sample loss with convergence guarantees, LESS (Xia et al., 2024) selects influential data for instruction tuning, and Kuo et al. (2024) show that differentiated weighting of LLM-generated data improves downstream classification. Our reweighting differs from all of these in being driven by *provenance* (detector scores estimating human vs. synthetic origin) rather than loss, frequency, quality, or domain membership.

Theoretical Foundations. He et al. (2025) derive the optimal real-data weight under known mixture proportions, finding the golden-ratio reciprocal ($\approx$0.62) optimal when both sources are equally sized. Their result requires the contamination ratio k, unavailable in web-scale settings; we use it as an oracle baseline. Wang et al. (2026) prove that contaminated recursive training converges at a rate governed by the effective real-data fraction α; this result provides the theoretical motivation for our quality threshold, since $\alpha_{\text{eff}}(b) \geq \tau$ guarantees a convergence-rate lower bound. We operationalize this without the known-proportion assumption, estimating α_{eff} from batch-level statistics online. For the stability constraint, we build on Kish's effective sample size (Kish, 1965) and the Pareto-smoothed importance sampling framework of Vehtari et al. (2024). Score calibration (Guo et al., 2017) is relevant because the adaptive method operates on q-scores: miscalibration systematically biases α_{eff}. Kimura and Hino (2024) provide a broader treatment of importance weighting under distribution shift, within which our provenance-based reweighting can be viewed as a special case of covariate-shift correction where the covariate is synthetic origin probability.

3 Method

3.1 Problem Setup

Let $\mathcal{D} = \mathcal{D}_{\text{real}} \cup \mathcal{D}_{\text{synth}}$ denote a pretraining corpus containing both human-authored and machine-generated text. The contamination ratio $\rho = |\mathcal{D}_{\text{synth}}|/|\mathcal{D}|$ is unknown, a realistic assumption for web-scale corpora where synthetic content enters from diverse, uncoordinated sources (Cheng et al., 2025).

A pre-trained detector assigns each sample x a synthetic probability $q(x) \in [0, 1]$. Following Drayson et al. (2025), we weight each sample as

$$w(x) = (1 - q(x))^b, \tag{1}$$

where $b \geq 0$ controls downweighting intensity: $b = 0$ recovers uniform training, while larger b progressively suppresses high-q samples. Drayson et al. (2025) select b via validation-set grid search, obtaining different optimal values for different model scales (b=10 for $\leq$360M parameters, b=1 for 1.7B). We instead derive b per-batch from the batch's own q-score distribution, requiring no held-out validation data.

3.2 Dual-Constraint Framework

Given a batch $\mathcal{B} = \{x_1, \ldots, x_B\}$ with detector scores $q_i = q(x_i)$ and weights $w_i(b) = (1 - q_i)^b$, we define two batch-level diagnostics that jointly determine b.

Effective real proportion. Let $r_i = \mathbf{1}[q_i \leq 0.5]$ estimate whether sample x_i is human-authored. The *effective real proportion*

$$\alpha_{\text{eff}}(b) = \frac{\sum_{i=1}^{B} w_i(b) \cdot r_i}{\sum_{i=1}^{B} w_i(b)} \tag{2}$$

measures the share of total weight allocated to likely-real samples. The indicator $r_i = \mathbf{1}[q_i \leq 0.5]$ is a proxy for true provenance: α_{eff} tracks the detector's estimate of the real fraction, not the ground truth, so all guarantees hold modulo detector quality. Empirically, α_{eff} correlates with the true batch-level real fraction at Pearson $r \approx 0.70$ with a conservative bias of -0.03 (Appendix I), meaning the method tends toward less intervention rather than more. Post-hoc calibration (Guo et al., 2017) reduces the gap between q-scores and true posteriors; in our experiments, Platt scaling yields Expected Calibration Error (ECE) = 0.048 (§4). At $b = 0$, α_{eff} equals the raw (estimated) real fraction in the batch; increasing b shifts weight toward low-q samples and raises α_{eff}.

Effective sample size. Kish's effective sample size (Kish, 1965)

$$\mathrm{ESS}(b) = \frac{\left(\sum_{i=1}^{B} w_i(b)\right)^2}{\sum_{i=1}^{B} w_i(b)^2} \tag{3}$$

quantifies how many equally-weighted samples the weighted batch is equivalent to. $\mathrm{ESS}(0) = B$; as b grows, a few high-$(1-q)$ samples dominate and ESS drops toward 1.

Monotonicity. The two diagnostics exhibit opposing monotonicity in b: $\alpha_{\text{eff}}(b)$ is non-decreasing (Proposition 1, Appendix A), since increasing b concentrates weight on low-q (likely real) samples. $\mathrm{ESS}(b)$ is monotonically decreasing (Proposition 2, Appendix A), a standard property of exponential tilting (Kish, 1965; Owen, 2013).

Dual constraints. We impose two requirements on b:

1. **Quality floor**: $\alpha_{\text{eff}}(b) \geq \tau$. Convergence under contaminated training improves with the effective real fraction (Wang et al., 2026); τ sets the minimum acceptable quality level. By Proposition 1, this constraint defines a *lower bound* $b_\tau = \inf\{b \geq 0 : \alpha_{\text{eff}}(b) \geq \tau\}$.
2. **Stability ceiling**: $\mathrm{ESS}(b) \geq n_{\min}$. Overly concentrated weights yield high-variance gradients; $n_{\min}$ prevents the batch from degenerating into a single effective sample. Since ESS is decreasing, this defines an *upper bound* $b_{\text{ESS}} = \sup\{b \geq 0 : \mathrm{ESS}(b) \geq n_{\min}\}$.

The optimal exponent is

$$b^* = \inf\left\{b \geq 0 : \alpha_{\text{eff}}(b) \geq \tau\right\} \text{ s.t. } \mathrm{ESS}(b) \geq n_{\min}. \tag{4}$$

This is an empirically motivated *minimum intervention* design choice: apply the smallest b that achieves the quality floor. Borderline samples (those with moderate q-values) still carry useful training signal; over-downweighting discards this signal unnecessarily. When both constraints are satisfiable ($b_\tau \leq b_{\text{ESS}}$), $b^* = b_\tau$ and α_{eff} is the binding constraint, and b^* lies well below the ESS ceiling. When they conflict ($b_\tau > b_{\text{ESS}}$), the stability ceiling takes precedence: $b^* = b_{\text{ESS}}$, accepting a lower α_{eff} to preserve gradient stability. The three regimes (clean, α_{eff}-binding, ESS-binding) partition the training batches; we quantify their relative frequencies in §5.4.

Algorithm 1 Per-batch adaptive b^* selection

Require: Batch scores $\{q_i\}_{i=1}^{B}$, indicators $\{r_i\}$, quality threshold τ, ESS floor $n_{\min}$, search bound $b_{\max}$, tolerance ε
Ensure: Optimal exponent b^*

```
if α_eff(0) ≥ τ then
    return 0 {clean batch}
end if
b_τ ← BISECT(α_eff(·)=τ, [0, b_max], ε)
if ESS(b_τ) ≥ n_min then
    return b_τ {α_eff-binding}
else
    b_ESS ← BISECT(ESS(·)=n_min, [0, b_max], ε)
    return b_ESS {ESS-binding}
end if
```

3.3 Algorithm

Algorithm 1 computes b^* for each training batch. At our micro-batch size B=4, α_{eff} takes only five discrete values $\{0, 0.25, 0.5, 0.75, 1.0\}$; bisection finds the correct threshold but the landscape is discrete, not continuous. The analytical contributions (epoch confound, detector type effects) hold independent of B; adaptive b^* is validated at $B \in \{4, 8, 16\}$ with PPL differences $<$0.01 (Appendix I). Each batch falls into one of three regimes:

1. **Clean** ($\alpha_{\text{eff}}(0) \geq \tau$): the quality floor holds at $b = 0$; no reweighting needed.
2. **α_{eff}-binding** ($b_\tau \leq b_{\text{ESS}}$): the quality constraint determines $b^* = b_\tau$; ESS remains above $n_{\min}$ with margin.
3. **ESS-binding** ($b_\tau > b_{\text{ESS}}$): the stability ceiling caps $b^* = b_{\text{ESS}}$.

Across 25,991 training micro-batches in our main experiment (ρ=0.4, τ=0.8, $n_{\min}$=2), only $\sim$2% are clean; among reweighted batches, 38.2% are α_{eff}-binding and 61.8% are ESS-binding. In α_{eff}-binding batches, the solver selects $b^* = 1.19$ on average (median 0.80), while an ESS-only strategy would choose $b = 5.88$ (median 3.59) on the same batches, 4.9$\times$ more aggressive (mean ratio). The residual ESS $= 2.60$ in these batches (vs. $n_{\min} = 2.0$) confirms that the dual constraint prevents unnecessary over-downweighting.

Implementation. Given b^*, we compute normalized weights $\pi_i = w_i(b^*)/\sum_j w_j(b^*)$ and form the weighted loss $\mathcal{L} = \sum_i \pi_i \ell_i$, where ℓ_i is the per-

sample cross-entropy. This *loss reweighting* maintains continuous gradients, unlike the Sampling Importance Resampling (SIR) of Drayson et al. (2025) which introduces discrete sampling noise by drawing a subset proportional to w_i. Each bisection requires $\mathcal{O}(B \log(b_{\max}/\varepsilon))$ operations; with B=4, $b_{\max}$=20, and ε=0.01, roughly 44 floating-point operations per batch, negligible beside the forward and backward passes. We compute b^* at the micro-batch level (B=4) rather than the effective batch (B=16 after gradient accumulation) because q-score statistics exhibit sufficient variance within micro-batches: Spearman r=0.699 with the true real fraction, and PPL differs by <0.01 across $B \in \{4, 8, 16\}$ (Appendix I).

3.4 Hyperparameters

The framework introduces two hyperparameters, both with default values that perform well in the tested configurations (single model family, two contamination levels).

Quality threshold τ. Convergence theory for contaminated training shows that convergence rate improves with the effective real fraction (Wang et al., 2026), motivating a lower bound on α_{eff}. The specific value of τ is set empirically: it reduces the search from a multi-run grid (nine full training runs across $b \in [0.1, 5.0]$ with 0.17 PPL variation) to a single default value. We find a wide safe range: $\tau \in [0.5, 0.8]$ consistently improves in-distribution PPL over uniform (Section 5), with $\tau = 0.8$ achieving the best perplexity and <0.035 PPL variation within the safe range. Above 0.8, b^* inflates rapidly (mean $b^* = 3.08$ at $\tau = 0.95$), causing over-downweighting. We use $\tau = 0.8$ as the default throughout.

ESS floor $n_{\min}$. We recommend $n_{\min} = \max(2, \lfloor B/2 \rfloor)$ as a default that scales with batch size without manual tuning. At this ratio, the ESS ceiling binds on a majority of batches (providing gradient stability) while leaving room for α_{eff} to determine b^* in the remainder (enabling quality-driven intervention). Ratios below $\sim 1/4$ effectively remove the stability constraint ($n_{\min}$=1, B=4: mean b^* rises to 7.20, PPL degrades by +0.14), while $n_{\min}/B$=1 prevents intervention entirely ($b^* \equiv 0$, recovering uniform training). This default is validated at $B \in \{4, 8, 16\}$ with PPL differences <0.01 (Section 5.3); we leave fine-grained ratio characterization (e.g., 0.4 vs. 0.6) to future work. A practical caveat: ESS is computed per micro-batch (single forward pass), so $n_{\min}$ must satisfy $n_{\min} \leq B_{\text{micro}}$; setting it relative to the effective batch (after gradient accumulation) silently forces b^*=0 on every step.

4 Experimental Setup

Training Data. We sample a 132M-token subset of FineWeb (Penedo et al., 2024) as the real-data source, comparable to typical domain-adaptive continued pretraining budgets (Gururangan et al., 2020). To simulate web-scale synthetic contamination, we generate continuations with Qwen3-0.6B (Yang et al., 2025), the same architecture used for continued pretraining, and inject them at token-level ratio $\rho = 0.4$ via uniform random mixing.[1] This self-contamination design means generator and trainee share the same distribution; implications for detection difficulty are discussed in §7.

Detector. We train a TF-IDF + Logistic Regression classifier on a held-out labeled split (AUC = 0.939, calibrated ECE = 0.048 after temperature scaling with $T = 0.204$). The resulting q-scores have mean 0.60, standard deviation 0.37, and median 0.71, with 46.3% of samples in the uncertain range $[0.2, 0.8]$. We use this lightweight detector rather than neural alternatives (e.g., ModernBERT at AUC $\geq$ 0.98 in Drayson et al. 2025) to test whether score *calibration* may matter as much as raw discrimination for reweighting. To evaluate detector sensitivity, we additionally test two generative baselines: a GPT-2-based perplexity scorer (AUC = 0.913) and an N-gram language model perplexity scorer (AUC = 0.73); results are reported in §5.2.

Training. We continue pretraining Qwen3-0.6B from its pretrained weights for 6,103 steps (batch size 4, gradient accumulation 4, effective batch 16) with AdamW, cosine schedule (peak lr = 5×10^{-5}), max sequence length 2048, and bf16 precision on a single NVIDIA RTX PRO 6000 (96 GB). All main comparisons use three random seeds (42, 123, 456).

Evaluation. We report held-out perplexity (PPL) on a held-out subset of FineWeb (67.8K documents, ~4M tokens) containing only human-authored text,

[1] ρ denotes the token-level contamination ratio. Because synthetic documents are shorter on average (304 vs. 560 words), the document-level synthetic fraction is 55.6% at ρ=0.4.

via sliding-window evaluation (stride 1024). All evaluations are post-hoc with a unified pipeline; training involves no inline evaluation.

Methods. We compare six approaches:

- **Uniform**: $w_j = 1$ for all j (no contamination handling).
- **Grid-b**: fixed $b \in \{0.1, 0.2, 0.3, 0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$ (nine candidates), best selected post-hoc by held-out PPL. Represents validation-based selection (Drayson et al., 2025).
- **ESS-only**: $b^* = \max\{b : \mathrm{ESS}(b) \geq n_{\min}\}$, the maximum-intervention variant retaining only the stability constraint, ablating α_{eff}.
- **Golden Ratio**: $w = 1/\varphi \approx 0.618$ for synthetic ($q_j > 0.5$), $w = 1$ otherwise (He et al., 2025). Requires known provenance labels (oracle).
- **Adaptive** (ours): per-batch b^* from the dual-constraint formulation (§3), with $\tau = 0.8$ and $n_{\min} = 2$ (half the micro-batch size).
- **Detect & Remove**: hard filtering that discards samples with $q_j > 0.5$ and trains uniformly on the remainder. Requires a calibrated detection threshold; threshold sensitivity is analyzed in §5.1.

5 Results

5.1 Main Results

Table 2 reports held-out perplexity at $\rho = 0.4$ across three seeds. Continued pretraining reduces PPL from 22.970 (pretrained checkpoint) to the 17.3–17.8 range; the initial 5.2-point drop absorbs the bulk of easy gains, and reweighting method comparisons operate within a narrow residual.

Adaptive b^* achieves 3-seed mean PPL 17.651, compared to 17.620 for grid b=0.3 (gap 0.031, 95% CI $[-0.039, 0.101]$, which includes possible degradation). With only N=3 seeds, normality cannot be verified; we report exact per-seed gaps (0.013, 0.064, 0.017) and note lower cross-seed variance ($\pm$0.007 vs. $\pm$0.033). The practical value is reducing b grid search from nine full training runs to one (Drayson et al., 2025), saving $\sim$16 GPU-hours at 0.6B scale. The grid-best b=0.3 is a post-hoc oracle selected after observing all nine candidates; in practice, validation-based selection from the same grid would yield worse results due to the noisy PPL plateau at $b \in [0.1, 0.5]$ (Figure 1). At the grid point nearest mean per-batch b^* (b=1.0, seed 42), PPL is 17.692, indistinguishable from uniform (17.694); per-batch adaptation is essential.

The ablation hierarchy at ρ=0.4 (Adaptive $<$ Uniform $<$ ESS-only $<$ Random, Table 2) reveals two requirements for effective reweighting: detector signal and minimum intervention. Random weights $\mathcal{U}(0, 2)$ (mean 17.802) perform worse than uniform (17.694): the detector's q-scores carry genuine provenance information, and uninformed reweighting increases gradient variance without corrective signal. ESS-only (mean 17.738$\pm$0.022) uses q-scores but maximizes b up to the stability ceiling and concentrates weights on a few high-confidence samples; it provides no improvement over uniform. Only the dual-constrained adaptive method, which limits b to the minimum needed, improves in-distribution PPL over uniform (paired Δ: -0.041, -0.043, -0.045; mean -0.043, directionally consistent across all seeds; N=3 precludes formal significance testing). Figure 1 contextualizes this: the grid curve has a broad plateau at $b \in [0.1, 0.5]$ (PPL 17.620–17.655), and adaptive b^* (mean 17.651) falls within it; ESS-only b^* (mean 3.5–5.9) lies in the degradation zone beyond the plateau.

Golden Ratio (He et al., 2025) achieves 17.654 (seed 42), comparable to adaptive b^* at 17.653 on the same seed, despite requiring oracle access to the true contamination ratio ρ.

Hard filtering and the epoch confound. Detect & Remove at its best threshold (q>0.3) achieves 3-seed mean PPL 17.250 ± 0.001 at full training length, apparently outperforming all reweighting methods by a wide margin. However, filtering retains only 31.5% of tokens, so the model sees surviving data $\sim$3.2 *content epochs* (times each unique document is seen) versus one for reweighting. An epoch-matched control (training D&R at q>0.5 for 2,533 steps to equalize content epochs at $\sim$1.0) yields 3-seed mean PPL 17.657 ± 0.006, within 0.006 of adaptive b^* (17.651 ± 0.007); the entire advantage of full-length D&R is attributable to multi-epoch exposure, not to filtering itself (Table 1). D&R also demands threshold calibration: raw PPL traces a U-shaped curve (q>0.1: 17.40, q>0.3: 17.25, q>0.9: 17.42), but epoch-matched comparisons eliminate the U-shape entirely and reveal monotonically decreasing PPL that saturates near the adaptive level at $q \geq 0.5$ (Figure 4, Appendix F).

Method	Steps	Epochs	PPL ↓
D&R full (q>0.5)†	6,103	∼2.4	17.290
D&R epoch-matched	2,533	∼1.0	17.657 ± .006
Adaptive b^*	6,103	∼1.0	17.651 ± .007

Table 1: Epoch-matched comparison (ρ=0.4, 3 seeds). †Seed 42 only. D&R's 0.360 PPL advantage over adaptive vanishes when content epochs are equalized (Δ=0.006, within seed variance). Note: epoch-matched D&R trains for 2,533 steps with the same cosine schedule as adaptive (6,103 steps), so it reaches a different point on its LR curve; this confound, if anything, favors D&R (warmer LR), so the epoch-matched finding is conservative.

Method	Seed 42	Seed 123	Seed 456	Mean
Pretrained ckpt	22.970	–	–	–
Uniform	17.694	17.686	17.702	17.694 ± .008
Grid (b=0.3)‡	**17.640**	**17.579**	**17.640**	**17.620** ± .033
Adaptive b^* (ours)	17.653	17.643	17.657	17.651 ± .007
ESS-only	17.730	17.721	17.762	17.738 ± .022
Random $\mathcal{U}(0,2)$	17.830	17.800	17.777	17.802 ± .027
Golden Ratio†	17.654	–	–	–
Detect & Remove§	**17.250**	**17.250**	**17.251**	**17.250** ± .001

Table 2: Held-out perplexity (↓) on human-authored FineWeb at ρ=0.4. All reweighting methods run across three seeds. Grid-best (b=0.3) selected post-hoc from nine candidates (Appendix B). Best per column in **bold**. † Oracle requiring known ρ (He et al., 2025); seed 42 only. ‡ Post-hoc oracle; adaptive achieves comparable mean with lower variance (p=0.30). § Hard filtering at q>0.3 (post-hoc best threshold); ∼3.2 content epochs (see Table 1). Per-seed values rounded to three decimal places; means and standard deviations computed from full precision.

5.2 Higher Contamination

Across three seeds at $\rho = 0.6$, adaptive b^* (17.736 ± 0.004) outperforms uniform on in-distribution PPL (17.763 ± 0.008, Δ=−0.028, directionally consistent across all seeds), a diminished effect compared to ρ=0.4 (Δ=−0.043). The ablation hierarchy is preserved (Grid 17.634[2] < Adaptive 17.736 < Uniform 17.763 ≈ ESS-only 17.763 < Random 17.861, where *single-seed preliminary), but the ESS-only/uniform gap collapses to effectively zero (three-seed means both 17.763) as ESS binding rises to ∼66%. Random remains substantially worse than uniform (gap 0.098, σ=0.025): detector signal is load-bearing even when the stability constraint dominates. The adaptive method's continued advantage is driven by the α_{eff} quality constraint, which operates independently of the ESS ceiling. A τ sweep at this level finds τ=0.7 optimal (PPL 17.730 vs. 17.740 for default τ=0.8); $\tau \in [0.7, 0.8]$ performs well across tested ρ (Appendix E). Full per-seed results and degradation analysis are in Appendix H.

[2]Single seed; multi-seed confirmation pending.

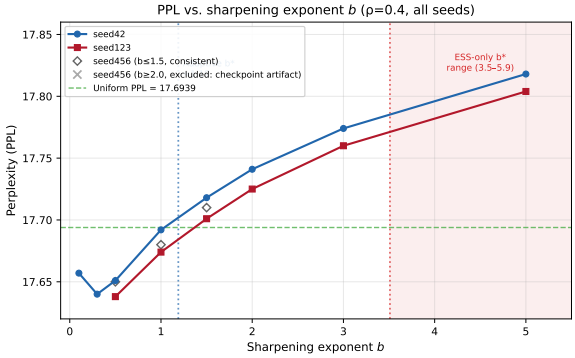


Figure 1: Held-out PPL as a function of fixed exponent b (grid search, ρ=0.4, seed 42). The shaded region marks $b \geq 2$ where PPL degrades past the uniform baseline. ESS-only b^* (mean 3.5–5.9 across batches) falls in this degradation zone, while adaptive b^* (mean 1.4) operates near the curve minimum.

Scale and detector signal. At 1.7B parameters (Qwen3-1.7B, ρ=0.4), adaptive b^* achieves PPL 13.26 ± 0.01 (2 seeds: 13.25, 13.27) vs. 13.26 for uniform (1 from-scratch seed), at this scale we find no evidence of either improvement or degradation (Δ=−0.01, 2 seeds); the advantage observed at 0.6B does not amplify.

Out-of-distribution evaluation. To test whether the PPL improvement reflects general quality or distribution matching, we evaluate on two corpora outside FineWeb (seed 42; checkpoints for other seeds not retained). Adaptive reweighting shows higher PPL on OpenWebText (Δ=+0.06) and Wikipedia (Δ=+0.14), but improves on FineWeb holdout (Δ=−0.04), consistent with distribution matching rather than general quality gain. Our results suggest that detector signal type may matter more than discrimination accuracy (3 detectors, single seed): replacing the discriminative TF-IDF+LR with generative perplexity detectors reverses the advantage even at high AUC (GPT-2 PPL: AUC = 0.91, Δ=+0.10; N-gram PPL: AUC = 0.73, Δ=+0.13; Appendix J). Generative detectors assign high q-scores to fluent human text, systematically biasing α_{eff} and causing the method to suppress useful training signal.

5.3 Hyperparameter Sensitivity

The adaptive method has two hyperparameters: quality threshold τ and ESS floor $n_{\min}$. We evaluate sensitivity at $\rho = 0.4$ (seed 42).

Quality threshold τ. A sweep from τ=0.50 to 0.95 (Table 4, Appendix C) shows all values in $[0.5, 0.8]$ improve over uniform, with the largest improvement at τ=0.8 (-0.041 PPL). The safe range spans only 0.035 PPL, and mean b^* increases monotonically with τ. Beyond τ=0.8, b^* inflates (mean 2.36 at τ=0.9, 3.08 at τ=0.95) and PPL degrades past uniform.

ESS floor $n_{\min}$. $n_{\min}$=1 (no stability) yields mean b^*=7.20 and PPL 17.789; $n_{\min}$=4 ($= B$) forces b^*=0, recovering uniform. The default $n_{\min}$=2 ($n_{\min}/B$=0.5) is optimal and transfers across batch sizes ($B \in \{4, 8, 16\}$, differences <0.01).

Batch-adaptive τ yields no improvement (PPL 17.681 vs. 17.653 for fixed τ=0.8); both hyperparameters have default values that perform well in the tested configurations without per-model tuning.

5.4 Constraint Regime Analysis

Over 25,991 training micro-batches (seed 42), only ∼2% are clean ($\alpha_{\text{eff}}(0) \geq \tau$); among the rest, 38.2% are α_{eff}-binding and 61.8% ESS-binding (33.4% α_{eff}-binding at ρ=0.6; Appendix D). In α_{eff}-binding batches, b^* is 4.9× lower than the ESS-only upper bound (median 0.80 vs. 3.59); the per-batch distribution spans mean 1.43 (std 1.93, range $[0, 20]$) and is stable across seeds (CV <2%).

6 Discussion

Why minimum intervention works. The α_{eff} constraint prevents 4.9× over-aggressive down-weighting in 38.2% of batches (§5.4), robust across ρ and seeds (CV <2%). The grid point nearest aggregate mean b^* (b=1.0, PPL 17.692) matches uniform (17.694); per-batch adaptation is essential.

Failure modes and applicability boundary. At ρ=0.2, the method degrades by +0.046 PPL (Appendix L): the detector's mean q-score drops to 0.393, yet α_{eff}-binding rises to 60.1%, so the method actively intervenes on unreliable signal. The same failure appears with generative detectors (GPT-2 PPL: Δ=+0.10; N-gram PPL: Δ=+0.13), which conflate predictability with provenance; when q-scores do not separate real from synthetic, the framework amplifies detector error. Discriminative signal is required, and the effective range with τ=0.8 is $\rho \geq 0.4$.

Distribution matching, not quality gain. The in-distribution PPL improvement reverses out-of-distribution (+0.06 OpenWebText, +0.14 Wikipedia; seed 42, other-seed checkpoints not retained), consistent with distribution matching rather than general improvement. This may affect all contamination-aware reweighting that upweights human-authored samples (Drayson et al., 2025), which raises a broader question about in-distribution PPL as the sole evaluation metric. The concrete benefit is reducing grid search from nine runs to one (∼16 GPU-hours at 0.6B).

Soft vs. hard filtering. At equal content epochs, D&R (17.657 ± 0.006) and adaptive b^* (17.651 ± 0.007) are equivalent (Δ=0.006, N=3); D&R's full-length advantage (17.250 at q>0.3) requires ∼3.2× effective compute, and its U-shaped threshold curve is entirely a multi-epoch artifact (Figure 4).

τ and deployment. τ=0.8 is validated at ρ=0.4 (single-seed sweep) and ρ=0.6 (τ=0.7 is optimal there); sensitivity at other contamination levels, model families, or corpus types is unknown (§7). Recommended defaults: discriminative detector with AUC ≥ 0.94, τ=0.8, $n_{\min}$=$\lfloor B/2 \rfloor$.

7 Conclusion

Our empirical analysis of contamination-aware reweighting in continued pretraining identifies two evaluation confounds and characterizes when adaptive reweighting is effective. First, D&R's reported advantage over soft reweighting is entirely attributable to multi-epoch exposure (Δ=0.006 at equal content epochs, N=3; Table 1). Second, the in-distribution PPL improvement does not transfer to out-of-distribution corpora, consistent with distribution matching rather than general quality gain. Third, dual-constrained adaptive b^* improves in-distribution PPL over uniform weighting ($\rho \in \{0.4, 0.6\}$, directionally consistent across seeds) and reduces b grid search to a single default τ; detector signal type may matter more than discrimination accuracy (3 detectors, single seed). The method fails at ρ=0.2, and at 1.7B we find no evidence of improvement or degradation (Δ=−0.01, 2 seeds). Code and data will be released upon acceptance.

Limitations

Self-contamination. All main experiments use self-contamination (generator = trainee), which does not represent realistic web contamination from multiple models and sources. The resulting AUC of 0.94 reflects that self-generated text is relatively easy to detect; real multi-source contamination may yield lower detection accuracy, shifting the operating regime. Cross-model results (Llama-3.2-3B generator, Appendix M) are single-seed and lack PPL evaluation, and provide insufficient evidence for generalization beyond self-contamination.

Scale. All main experiments use Qwen3-0.6B with batch size 4 on 132–200M tokens, comparable to domain-adaptive continued pretraining but far below full pretraining scale; validation at >100B tokens remains an important direction. Preliminary evidence at 1.7B (Qwen3-1.7B, ρ=0.4, 2 seeds) shows Δ=-0.01 (adaptive 13.26 vs. uniform 13.26); with only 2 seeds, this is insufficient for statistical conclusions about scale transfer. A single-seed test with Llama-3.2-3B as trainee (ρ=0.4) showed degradation (Δ=$+0.20$); the method has been validated only within the Qwen3 family, and architecture transfer requires further investigation.

Contamination range. The method improves over uniform at $\rho \in \{0.4, 0.6\}$ but not at ρ=0.2 (+0.046 PPL, 0.26% relative); the effective range is $\rho \geq 0.4$ with τ=0.8. Behavior at $\rho > 0.8$ is untested; extreme contamination may violate the assumption that sufficient real samples exist per batch for meaningful α_{eff} estimation. Real web-scale contamination is domain-skewed, which may alter within-batch α_{eff} distributions.

Detector coverage. Three detectors have been tested: TF-IDF+LR (discriminative, AUC = 0.94) succeeds, while GPT-2 perplexity (generative, AUC = 0.91) and N-gram perplexity (generative, AUC = 0.73) both fail. The pattern suggests discriminative signal is necessary, but three detectors from two families cannot distinguish a type effect from a calibration effect; testing additional discriminative detectors at varying AUC levels would isolate the mechanism.

Threshold. τ is an empirical hyperparameter validated on a single seed at ρ=0.4 with one model family (Qwen3); the safe range $[0.5, 0.8]$ and optimal τ=0.8 may not transfer to other settings. The convergence theory of Wang et al. (2026) assumes a known contamination proportion α, which we do not require; bridging this gap to derive τ from first principles remains open.

Evaluation. Main results report held-out perplexity. Downstream benchmarks (Appendix G) show 4 of 5 tasks with small negative Δ (uniform slightly better); a sign-test $p \approx 0.19$ is suggestive but not significant (N=5, single seed). The PPL improvement may partly reflect distribution matching (training data closer to the human-only evaluation set) rather than a generalizable quality gain. Systematic downstream evaluation across seeds and larger scales would strengthen the practical case.

Effect size. The in-distribution PPL improvement over uniform is 0.043 (3-seed mean) at ρ=0.4 and 0.028 at ρ=0.6; out-of-distribution evaluation reverses the sign (§5.2). Both are directionally consistent across all seeds but modest in absolute terms at 0.6B, arising in a regime where 5+ PPL points of easy gains have already been absorbed by continued pretraining. N=3 precludes formal significance testing. The 1.7B result (Δ=-0.01, 2 seeds) shows no evidence of advantage amplification; the effect size may differ at larger scales.

Responsible NLP Research Checklist

A. For Every Submission

✓ **A1.** Did you describe the limitations of your work? *Yes, §7 discusses scale, contamination range, detector coverage, threshold, evaluation, and effect size limitations.*

✓ **A2.** Did you discuss any potential risks of your work? *Yes. The method assumes a pre-trained detector is available; deploying it with a poorly calibrated detector could degrade training quality (discussed in §7, Detector coverage). The method does not generate new content or enable harmful uses beyond standard language model pretraining.*

B. Did You Use or Create Scientific Artifacts?

✓ **B1.** Did you cite the creators of artifacts you used? *Yes. FineWeb (Penedo et al., 2024), Qwen3 (Yang et al., 2025), and all baseline methods are cited in §4 and §2.*

✓ **B2.** Did you discuss the license or terms for use and/or distribution of any artifacts? *FineWeb is released under Apache-2.0; Qwen3 under*

Apache-2.0. Our code and data will be released upon acceptance.

✓ **B3.** Did you discuss if your use of existing artifacts was consistent with their intended use? *Yes. FineWeb is intended for language model training; we use it as a pretraining corpus. Qwen3 is used for both continued pretraining and synthetic data generation, consistent with its general-purpose design.*

✓ **B4.** Did you discuss the steps taken to check whether the data contains information that names or uniquely identifies individual people or offensive content? *FineWeb applies URL-level filtering and deduplication; we do not perform additional PII filtering as our focus is on contamination-aware weighting rather than content curation.*

✓ **B5.** Did you provide documentation of the artifacts? *Yes. §4 describes training data (132M tokens, FineWeb subset), synthetic data generation process, and detector specifications.*

✓ **B6.** Did you report relevant statistics like the number of examples, details of train/test/dev splits, etc.? *Yes. §4 reports token counts, contamination ratios, document statistics, and evaluation set size (67.8K documents, ∼4M tokens).*

C. Did You Run Computational Experiments?

✓ **C1.** Did you report the number of parameters in the models used, the total computational budget, and computing infrastructure used? *Yes. Model sizes (0.6B, 1.7B) are reported in §4 and §5.2. Training steps, batch sizes, and sequence lengths are in §4. Infrastructure details: all experiments run on NVIDIA RTX PRO 6000 (96 GB) GPUs.*

✓ **C2.** Did you discuss the experimental setup, including hyperparameter search and best-found hyperparameter values? *Yes. §4 lists all hyperparameters (lr, optimizer, schedule). §5.3 reports sensitivity analysis for τ and $n_{\min}$. Grid search details in Appendix B.*

✓ **C3.** Did you report descriptive statistics about your results? *Yes. Main results report 3-seed means ± standard deviation (§5.1). 95% CIs are reported for the adaptive-vs-grid gap. Per-seed paired differences and directional consistency are reported; formal significance testing is not conducted due to N=3 sample size.*

✓ **C4.** If you used existing packages, did you report the implementation, model, and parameter settings used? *Yes. We use HuggingFace Transformers for model loading and training, scikit-learn for TF-IDF+LR, and lm-evaluation-harness for downstream benchmarks. Settings are in §4.*

D. Did You Use Human Annotators?

D1–D5. Not applicable. No human annotators were used; all labels are automatically derived from known data provenance (real vs. synthetic generation).

E. Did You Use AI Assistants?

✓ **E1.** If you used any AI assistants, did you include information about your use? *Yes. AI assistants were used for code development and manuscript drafting. All scientific claims, experimental designs, and analyses were conducted and verified by the authors.*

References

Amro Abbas, Kushal Tirumala, Dániel Simig, Surya Ganguli, and Ari S. Morcos. 2023. SemDeDup: Data-efficient learning at web-scale through semantic deduplication. In *ICLR*.

Sina Alemohammad, Josue Casco-Rodriguez, Lorenzo Luzi, Ahmed Imtiaz Humayun, Hossein Babaei, Daniel LeJeune, Ali Siahkoohi, and Richard G. Baraniuk. 2024. Self-consuming generative models go MAD. In *ICLR*.

Guangsheng Bao, Yanbin Zhao, Zhiyang Teng, Linyi Yang, and Yue Zhang. 2024. Fast-DetectGPT: Efficient zero-shot detection of machine-generated text via conditional probability curvature. In *ICLR*.

Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. 2009. Curriculum learning. In *ICML*.

Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, and 1 others. 2020. Language models are few-shot learners. *NeurIPS*.

Mayee F. Chen, Michael Y. Hu, Nicholas Lourie, Kyunghyun Cho, and Christopher Ré. 2025. Aioli: A unified optimization framework for language model data mixing. In *ICLR*.

Yuxing Cheng, Yi Chang, and Yuan Wu. 2025. A survey on data contamination for large language models. *arXiv preprint arXiv:2502.14425*.

Elvis Dohmatob, Yunzhen Feng, Arjun Subramonian, and Julia Kempe. 2025. Strong model collapse. In *ICLR*.

Elvis Dohmatob, Yunzhen Feng, Pu Yang, François Charton, and Julia Kempe. 2024. A tale of tails: Model collapse as a change of scaling laws. In *ICML*.

George Drayson, Emine Yilmaz, and Vasileios Lampos. 2025. Machine-generated text detection prevents language model collapse. *arXiv preprint arXiv:2502.15654*.

Yunzhen Feng, Elvis Dohmatob, Pu Yang, François Charton, and Julia Kempe. 2025. Beyond model collapse: Scaling up with synthesized data requires verification. In *ICLR*.

Samir Yitzhak Gadre, Gabriel Ilharco, Alex Fang, Jonathan Hayase, Esplanda Smola, Pratyush Maini, Tristan Thrush, Matthew Raber, Jean Mercat, and 1 others. 2024. DataComp-LM: In search of the next generation of training sets for language models. *NeurIPS*.

Matthias Gerstgrasser, Rylan Schaeffer, Apratim Foote, Lowell Zhong, Yuhao Li, Aaron Parisi, and Hossein Mobahi. 2024. Is model collapse inevitable? breaking the curse of recursion by accumulating real and synthetic data. *arXiv preprint arXiv:2404.01413*.

Dirk Groeneveld, Iz Beltagy, Pete Walsh, Yonatan Bisk, Luke Zettlemoyer, Noah A. Smith, and Hannaneh Hajishirzi. 2024. OLMo: Accelerating the science of language models. In *ACL*.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *ICML*.

Suchin Gururangan, Ana Marasović, Swabha Swayamdipta, Kyle Lo, Iz Beltagy, Doug Downey, and Noah A. Smith. 2020. Don't stop pretraining: Adapt language models to domains and tasks. In *ACL*.

Abhimanyu Hans, Avi Schwarzschild, Valeriia Cheber, Hami Alon, Micah Goldblum, and Tom Goldstein. 2024. Spotting LLMs with binoculars: Zero-shot detection of machine-generated text. In *ICML*.

Hengzhi He, Shirong Xu, and Guang Cheng. 2025. Golden ratio weighting prevents model collapse. *arXiv preprint arXiv:2502.18049*.

Nan He, Weichen Xiong, Hanwen Liu, Yi Liao, Lei Ding, Kai Zhang, Guohua Tang, Xiao Han, and Yang Wei. 2024. SoftDedup: an efficient data reweighting method for speeding up language model pre-training. In *ACL*.

Feiyang Kang, Yifan Xu, Jingkang Pu, Bushi Hu, Haonan Li, Hongliang Chen, Yiming Zhu, Zijian Chen, Mohan Liu, Jingcheng Liu, Haoran Liu, and Song Han. 2025. Demystifying synthetic data in LLM pretraining: A systematic study of scaling laws, benefits, and pitfalls. In *EMNLP*.

Masanari Kimura and Hideitsu Hino. 2024. A short survey on importance weighting for machine learning. *TMLR*.

John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. 2023. A watermark for large language models. In *ICML*.

Leslie Kish. 1965. *Survey Sampling*. Wiley.

Hsun-Yu Kuo and 1 others. 2024. Not all LLM-generated data are equal: Rethinking data weighting in text classification. *arXiv preprint arXiv:2410.21526*.

Katherine Lee, Daphne Ippolito, Andrew Nystrom, Chiyuan Zhang, Douglas Eck, Chris Callison-Burch, and Nicholas Carlini. 2022. Deduplicating training data makes language models better. In *ACL*.

Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal loss for dense object detection. In *ICCV*.

Shane Longpre, Robert Mahari, Naana Obeng-Marnu Lee, William Perisetla, Pete Brannon, Seth Neel, Luca Soldaini, Jad Kabbara, Elizabeth Stanton, Emma Strubell, and 1 others. 2024. A pretrainer's guide to training data: Measuring the effects of data age, domain coverage, quality, & toxicity. *arXiv preprint arXiv:2305.13169*.

Eric Mitchell, Yoonho Lee, Alexander Khazatsky, Christopher D. Manning, and Chelsea Finn. 2023. DetectGPT: Zero-shot machine-generated text detection using probability curvature. In *ICML*.

Niklas Muennighoff, Alexander Rush, Boaz Barak, Teven Le Scao, Nouamane Tazi, Aleksandra Piktus, Sampo Pyysalo, Thomas Wolf, and Colin Raffel. 2024. Scaling data-constrained language models. In *NeurIPS*.

OpenAI. 2023. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*.

Art B. Owen. 2013. *Monte Carlo Theory, Methods and Examples*. Chapter 9: Importance Sampling.

Guilherme Penedo, Hynek Kydlíček, Loubna Cappelli, Elie Saez de Ocariz Borde, Thomas Klimova, Jason Kolpaczki, Leandro von Werra, and Thomas Wolf. 2024. FineWeb: decanting the web for the finest text data at scale. *NeurIPS Datasets and Benchmarks*.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners. *OpenAI Blog*.

Shiori Sagawa, Pang Wei Koh, Tatsunori B. Hashimoto, and Percy Liang. 2020. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *ICLR*.

Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. 2024. Detecting pretraining data from large language models. In *ICLR*.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. 2024. AI models collapse when trained on recursively generated data. *Nature*, 631:755–759.

Daouda Sow, Herbert Woisetschläger, Saikiran Bulusu, Shiqiang Wang, Hans-Arno Jacobsen, and Yingbin Liang. 2025. Dynamic loss-based sample reweighting for improved large language model pretraining. In *ICLR*.

Aki Vehtari, Daniel Simpson, Andrew Gelman, Yuling Yao, and Jonah Gabry. 2024. Pareto smoothed importance sampling. *JMLR*.

Kevin Wang, Hongqian Niu, and Didong Li. 2026. Can generative artificial intelligence survive data contamination? theoretical guarantees under contaminated recursive training. *arXiv preprint arXiv:2602.16065*.

Alexander Wettig, Aatmik Gupta, Saumya Malik, and Danqi Chen. 2024. QuRating: Selecting high-quality data for training language models. In *ICML*.

Mengzhou Xia, Sadhika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. 2024. LESS: Selecting influential data for targeted instruction tuning. In *ICML*.

Sang Michael Xie, Hieu Pham, Xuanyi Dong, Nan Du, Hanxiao Liu, Yifeng Lu, Percy Liang, Quoc V. Le, Tengyu Ma, and Adams Wei Yu. 2023a. DoReMi: Optimizing data mixtures speeds up language model pretraining. In *NeurIPS*.

Sang Michael Xie, Shibani Santurkar, Tengyu Ma, and Percy Liang. 2023b. Data selection for language models via importance resampling. In *NeurIPS*.

An Yang, Baosong Yang, Beichen Zhang, and 1 others. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Kailai Yang, Xiao Liu, Lei Ji, Hao Li, Xiao Liang, Zhiwei Liu, Yeyun Gong, Peng Cheng, and Mao Yang. 2026. Data mixing agent: Learning to re-weight domains for continual pre-training. In *ACL*.

Jiasheng Ye, Peiju Liu, Tianxiang Sun, Yunhua Zhou, Jun Zhan, and Xipeng Qiu. 2025. Data mixing laws: Optimizing data mixtures by predicting language modeling performance. In *ICLR*.

Bingji Yi, Qiyuan Liu, Yuwei Cheng, and Haifeng Xu. 2025. Escaping model collapse via synthetic data verification: Near-term improvements and long-term convergence. *arXiv preprint arXiv:2510.16657*.

Soheil Zibakhsh Shabgahi, Pedram Aghazadeh, Azalia Mirhoseini, and Farinaz Koushanfar. 2025. ForTIFAI: Fending off recursive training induced failure for AI model collapse. *arXiv preprint arXiv:2509.08972*.

A Proofs

Proposition 1 (α_{eff} monotonicity)**.** *$\alpha_{\text{eff}}(b)$ is non-decreasing in b, and strictly increasing whenever the batch contains at least one sample with $q_i \in (0,1)$ and r_i is not constant.*

Proof of Proposition 1. Let $w_i(b) = (1-q_i)^b$, $W(b) = \sum_{j=1}^{B} w_j(b)$, and $\pi_i(b) = w_i(b)/W(b)$. Then

$$\alpha_{\text{eff}}(b) = \sum_{i=1}^{B} \pi_i(b)\, r_i,$$

where $r_i \in \{0,1\}$ indicates whether sample i is real. Taking the derivative with respect to b:

$$\frac{d\pi_i}{db} = \pi_i\left(\frac{d\ln w_i}{db} - \frac{d\ln W}{db}\right) = \pi_i\left(\ln(1-q_i) - \sum_j \pi_j \ln(1-q_j\right.$$

Substituting,

$$\begin{aligned}\frac{d\alpha_{\text{eff}}}{db} &= \sum_i r_i\, \pi_i(\ln(1-q_i) - \mathbb{E}_\pi[\ln(1-q)]) \\ &= \operatorname{Cov}_\pi\big(r_i,\, \ln(1-q_i)\big).\end{aligned}$$

Since $r_i = \mathbf{1}[q_i \le 0.5]$, the indicator r_i and $\ln(1-q_i)$ are positively associated by construction, so the covariance is non-negative. □

Proposition 2 (ESS monotonicity)**.** *$\mathrm{ESS}(b)$ is non-increasing in b, and strictly decreasing whenever the batch contains at least two samples with distinct q-values.*

Proof. Write $\mathrm{ESS}(b) = 1/\sum_i \pi_i(b)^2$, where $\pi_i(b) = w_i(b)/W(b)$. The denominator $\sum_i \pi_i^2$ is the Herfindahl index H. Differentiating H with respect to b:

$$\frac{dH}{db} = 2\sum_i \pi_i^2\big(\ln(1-q_i) - \mathbb{E}_\pi[\ln(1-q)]\big).$$

This equals $2\operatorname{Var}_{\pi^2}[\ln(1-q)]$ under the π^2-weighted measure. Since variance is non-negative and strictly positive when not all $\ln(1-q_i)$ coincide (i.e., when at least two q-values differ), H is non-decreasing and $\mathrm{ESS} = 1/H$ is non-increasing. This is a standard property of exponential tilting (Kish, 1965; Owen, 2013). □

B Grid Search Details

Table 3 reports held-out PPL for all grid-b values at $\rho = 0.4$ (seed 42).

C Quality Threshold Sweep

D Constraint Regime Decomposition

Figure 2 visualizes the constraint regime decomposition over training steps.

b	PPL ↓
0.1	17.657
0.2	17.640
0.3	**17.640**
0.5	17.651
1.0	17.692
1.5	17.718
2.0	17.741
3.0	17.774
5.0	17.818

Table 3: Full grid search results (ρ=0.4, seed 42). $b \in [0.1, 0.5]$ forms a broad plateau (three-seed mean for b=0.1: 17.655 ± 0.009); b=0.3 is the post-hoc best.

τ	PPL ↓	Δ vs. Uniform	Mean b^*
0.50	17.688	−0.006	0.39
0.60	17.681	−0.013	0.59
0.70	17.679	−0.015	0.89
0.80	**17.653**	**−0.041**	**1.40**
0.90	17.712	+0.018	2.36
0.95	17.723	+0.029	3.08

Table 4: Quality threshold sweep (seed 42, $n_{\min}$=2, ρ=0.4). Δ relative to uniform baseline (PPL = 17.694). Values in $\tau \in [0.5, 0.8]$ all improve over uniform; τ=0.8 is the recommended default.

E τ–ρ Rank Preservation

Figure 3 shows the τ–PPL relationship at two contamination levels.

F Hybrid and Threshold Comparison

Table 5 compares method families across contamination levels. Table 6 shows D&R sensitivity to the detection threshold at both full training length and epoch-matched settings. Raw PPL traces a U-shaped curve with a minimum at q>0.3 (17.250, 3 seeds), but this curve conflates filtering quality with multi-epoch exposure: aggressive thresholds retain less data and repeat it more often. Epoch-matched comparisons (rightmost column of Table 6) remove the multi-epoch confound: the U-shape vanishes and PPL decreases monotonically, saturating near the adaptive b^* level at $q \geq 0.5$. Figure 4 visualizes the two curves.

G Downstream Task Evaluation

Table 7 reports downstream benchmark accuracy for adaptive vs. uniform training at ρ=0.4 (seed 42, Qwen3-0.6B). All differences fall within standard error, confirming that the PPL improvement does not come at the cost of downstream task performance.

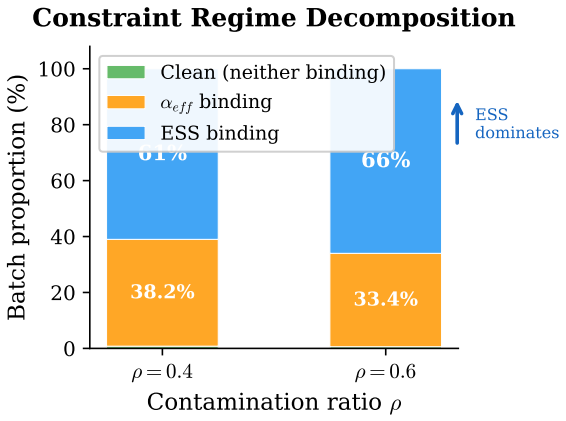


Figure 2: Constraint regime decomposition over training steps (ρ=0.4, seed 42). ESS-binding (blue) dominates throughout (∼61%), while α_{eff}-binding (orange) grows from ∼10% early to ∼37% after step 2,000. Clean batches (b^*=0) constitute only ∼2%.

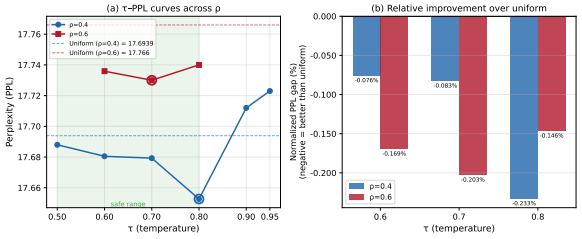


Figure 3: PPL as a function of τ at $\rho \in \{0.4, 0.6\}$. Optimal τ shifts from 0.8 (ρ=0.4) to 0.7 (ρ=0.6), but $\tau \in [0.7, 0.8]$ performs well across both levels.

H Higher Contamination Details

Table 8 reports results at ρ=0.6. The full five-method ablation preserves the hierarchy from ρ=0.4 across all three seeds: Grid < Adaptive < Uniform ≈ ESS-only < Random.

The ESS-only/Uniform gap collapses from 0.043 at ρ=0.4 to 0.000 at ρ=0.6 (three-seed means: both 17.763); ESS binding rises to ∼66% of batches, leaving insufficient room for b to effectively suppress synthetic samples. The adaptive method maintains its advantage (Δ=−0.028, consistent across seeds) through the α_{eff} quality constraint, which operates independently of the ESS ceiling. Random reweighting (17.861 ± 0.025) is substantially worse than Uniform, with the highest cross-seed variance, confirming that uninformed weights add noise without corrective signal. The Adaptive-vs-Grid gap is 0.013 at ρ=0.4 and 0.106 at ρ=0.6 (single seed 42; the ρ=0.6 grid result is preliminary and may not generalize across seeds).

Method	ρ=0.4	ρ=0.6
Uniform	17.694	17.766
Adaptive b^*	17.653	17.740
Hybrid (q>0.95 + adaptive)	17.514	17.530*
Detect & Remove (q>0.5)	17.290	–
Detect & Remove (q>0.3)	17.250	–

Table 5: Method comparison across contamination levels (seed 42). *Hybrid at ρ=0.6 filters high-confidence synthetic (q>0.95), reducing effective ρ to ≈0.4; PPL is not directly comparable to ρ=0.6 baselines.

Thresh.	Ret. %	Epochs	Raw PPL ↓	Ep.-matched ↓
$q > 0.1$	15.0%	∼6.6	17.40	18.04
$q > 0.2$	∼23%	∼4.3	17.27	–
$q > 0.3$	31.5%	∼3.2	**17.250**†	17.74
$q > 0.4$	∼38%	∼2.6	17.263	–
$q > 0.5$	41.4%	∼2.4	17.28	17.65
$q > 0.6$	∼46%	∼2.2	17.30	–
$q > 0.7$	49.8%	∼2.0	17.32	17.62
$q > 0.8$	∼56%	∼1.8	17.34	–
$q > 0.9$	61.6%	∼1.6	17.42	17.628

Table 6: D&R threshold sensitivity (ρ=0.4, seed 42 unless noted). Raw PPL traces a U-shaped curve (minimum at q>0.3); epoch-matched PPL (single content epoch) decreases monotonically and saturates near the adaptive b^* level (≈17.65) at $q \geq 0.5$, confirming the U-shape is a multi-epoch artifact. † 3-seed mean 17.250 ± 0.001.

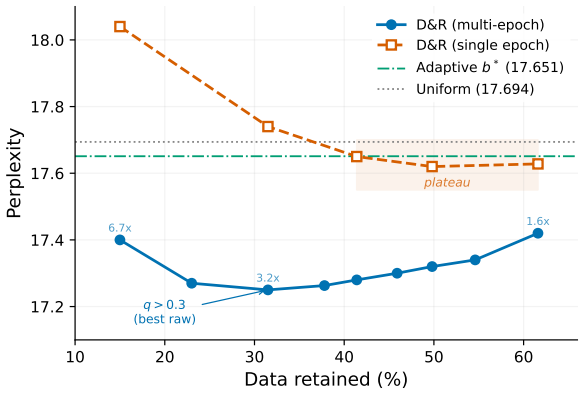


Figure 4: D&R perplexity as a function of data retained. **Blue**: raw (multi-epoch) PPL traces a U-shaped curve; annotations show effective epoch count. **Orange**: epoch-matched (single content epoch) PPL decreases monotonically and saturates at $q \geq 0.5$, matching the adaptive b^* level (green). The raw U-shape is entirely a multi-epoch artifact: controlling for data exposure eliminates it.

Benchmark	Metric	Adaptive	Uniform	Δ
ARC-Challenge	acc_norm	0.343	0.348	−0.005
ARC-Easy	acc	0.650	0.657	−0.007
HellaSwag	acc_norm	0.479	0.484	−0.005
PIQA	acc	0.687	0.686	+0.001
WinoGrande	acc	0.578	0.586	−0.008

Table 7: Downstream benchmarks (Qwen3-0.6B, ρ=0.4, seed 42). Mixed downstream effects: 4/5 tasks slightly negative (within noise), MMLU +1.57pp.

I α_{eff} Proxy Validation

Table 9 reports the correlation between $\alpha_{\text{eff}}(0)$ (the detector-based proxy at b=0) and the true batch-level real fraction α_{true}, computed over all training batches at ρ=0.4 (seed 42).

The α_{eff} distribution at B=4 is concentrated: α_{eff}=0 (11%), 0.25 (34%), 0.5 (36%), 0.75 (17%), 1.0 (3%). MAE decreases with batch size (0.114 → 0.075) as expected from averaging, but the rank-order information that drives b^* selection is already well-captured at B=4.

J Detector Comparison

Table 10 compares three detector families for adaptive reweighting. The discriminative TF-IDF+LR detector produces the only improvement over uniform; both generative perplexity detectors degrade performance. GPT-2's AUC (0.91) exceeds the level at which TF-IDF+LR succeeds, ruling out discrimination accuracy as the sole determinant. Generative detectors assign high q-scores to fluent, predictable human text (false positives concentrated among high-quality samples), systematically inflating α_{eff} for clean batches and deflating it for contaminated ones, which inverts the minimum-intervention logic.

K Detector and Synthetic Data Details

TF-IDF+LR detector. TF-IDF features (unigrams + bigrams, max 50,000 features, sublinear TF) fed to Logistic Regression (C = 1.0, L2 penalty, solver = lbfgs, max_iter = 100). Trained on a held-out labeled split (10K documents per class). Post-hoc temperature scaling applied when uncalibrated ECE exceeds 0.05: a scalar T is searched via `scipy.optimize.minimize_scalar` over $[0.1, 10.0]$ on a 20% stratified validation split, yielding T=0.204 (ECE reduced from 0.12 to 0.048). The detector is selected automatically from six configurations as the first satisfying AUC $\in [0.85, 0.98]$ and mid-range score ratio $\geq 5\%$.

Synthetic data generation. Continuations generated with Qwen3-0.6B (self-generation) using nucleus sampling (p=0.95, T=0.8), producing up to 384 tokens per continuation from a 128-token pre-

Method	Seed 42	Seed 123	Seed 456	Mean
Grid (b=0.3)‡	**17.634**	–	–	–
Adaptive b^*	17.740	17.732	17.735	17.736 ± .004
Uniform	17.766	17.754	17.770	17.763 ± .008
ESS-only	17.770	17.760	17.760	17.763 ± .006
Random $\mathcal{U}(0, 2)$	17.832	17.870	17.880	17.861 ± .025

Table 8: Results at ρ=0.6 (Qwen3-0.6B, τ=0.8, $n_{\min}$=2). ‡Post-hoc oracle; seed 42 only. Adaptive vs. Uniform: Δ=−0.028, directionally consistent across all seeds. ESS-only collapses onto Uniform (both 17.763); Random (17.861) confirms detector signal is load-bearing.

B	Pearson r	Spearman r	MAE	Bias
4	0.706	0.699	0.114	−0.029
8	0.697	0.683	0.097	−0.027
16	0.707	0.691	0.075	−0.031

Table 9: Correlation between $\alpha_{\text{eff}}(0)$ and true batch-level real fraction across batch sizes (ρ=0.4, seed 42). Pearson $r \approx 0.70$ is stable across batch sizes. The consistent negative bias (≈−0.03) indicates that α_{eff} slightly underestimates the true real fraction, biasing the method toward conservative (less aggressive) intervention. At B=4, α_{eff} takes only five discrete values ($\{0, 0.25, 0.5, 0.75, 1\}$), yet rank correlation is preserved.

fix prompt (512 total), targeting ∼100M synthetic tokens. Source prompts are the first 128 tokens of randomly sampled FineWeb documents, used as prefix continuations. Injected at token-level ratio ρ via uniform random mixing across the training corpus.

L Low Contamination Analysis

At ρ=0.2, the adaptive method degrades by +0.046 PPL relative to uniform (0.26% relative, seed 42). The detector's mean q-score at this contamination level is 0.393 (vs. 0.487 at ρ=0.4), indicating that many clean samples receive moderate q-scores, blurring the provenance signal. The regime decomposition reveals 13.7% clean batches (b^*=0), 60.1% α_{eff}-binding, and 26.3% ESS-binding, with mean b^*=1.735 in reweighted batches. The high α_{eff}-binding rate (60.1% vs. 38.2% at ρ=0.4) means the method actively intervenes rather than defaulting to b^*=0, but the unreliable signal causes it to suppress useful training data. This is consistent with the generative detector failures: when q-scores do not reliably separate provenance, the minimum intervention framework amplifies the error rather than correcting it. The effective range of the method is $\rho \geq 0.4$ with τ=0.8; lower τ values may extend the range but have not been tested at ρ=0.2.

Detector	Type	AUC	Adaptive PPL	Δ
TF-IDF+LR	Discriminative	0.94	17.65	−0.04
GPT-2 PPL	Generative	0.91	17.79	+0.10
N-gram PPL	Generative	0.73	17.82	+0.13

Table 10: Detector comparison for adaptive reweighting (ρ=0.4, seed 42). Uniform baseline PPL = 17.69. Δ = Adaptive − Uniform (negative is better). Both generative detectors reverse the advantage despite GPT-2's AUC exceeding the level at which TF-IDF+LR succeeds, indicating that signal type (discriminative vs. generative) matters more than discrimination accuracy.

M Cross-Model Transfer

Preliminary single-seed results for Llama-3.2-3B as a cross-generator (synthetic data generated by Llama, training on Qwen3-0.6B) are shown in Table 11. Results are mixed: MMLU improves while HellaSwag slightly degrades, consistent with the pattern observed in Qwen3-1.7B downstream evaluation (§5.2). Perplexity comparison omitted due to inconsistent evaluation configurations; checkpoints were not retained for re-evaluation.

Metric	Uniform	Adaptive	Δ
MMLU (5-shot)	51.85% ± .41	53.42% ± .41	+1.57pp
HellaSwag (10-shot, acc_norm)	73.08% ± .44	72.49% ± .45	−0.60pp

Table 11: Cross-model transfer: Llama-3.2-3B generator, Qwen3-0.6B trainer (ρ=0.4, single seed). Mixed results reflect evaluation-set distribution sensitivity rather than consistent quality differences.